



DEALER BEST PRACTICE SERIES

Parts Blasting
Enhancement

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1.0 Introduction

In a CRC, parts cleaning is typically performed in dedicated general parts cleaning work bay. Parts cleaning may utilize several technologies, including high-pressure cabinet washing, manual solvent washing, or gasket scraping. Parts cleaning must be managed to provide high quality rebuilds while minimizing costs and “over-cleaning”.

Clean parts introduce minimal contaminants into a component, reducing internal parts wear and providing longer service life. Clean parts also help assure that joints fit accurately and securely without future leaks. Worn parts often retain gasket material, hard carbon, and rust in their joint mating areas, even after high-pressure cabinet washing. These surface areas minimally require scraping and possibly buffing/blasting if the contamination is aggressive.

Many cleaning processes create airborne debris. Minimal effects find the debris deposited on staged clean parts and inside equipment, while a worst case could send debris through the entire shop rebuild/assembly areas. In all instances, uncontrolled airborne debris creates an adverse situation for repair quality.

This Best Practice describes a well-designed method of controlling the airborne debris generated when blast-cleaning parts. Methods in the scope include sand, glass-bead, and steel-shot blasting. A segregated blast room is used with blast cabinets, often considered a standard piece of CRC cleaning equipment, to help control blasting emissions. Buffers are often used in the general cleaning area but tough cleaning is often performed more effectively with blast cabinet cleaners.

2.0 Best Practice Description

Aggressive part cleaning is often performed with a handheld angle buffers over a downdraft table or bench-mounted buffer. A cabinet-type blast cleaner is typically a more efficient (faster) tool if the piece part has a sensitive surface, and/or is heavily corroded or carbon coated. Controlling the blast cabinet’s airborne debris is often difficult. Most cabinets leak some abrasive dust when new, and get progressively worse with age. Additionally, controlling airborne debris from bench grinders is difficult since the blast cabinet seals and extraction/separation system typically emit debris over a large space while running.

The most effective debris control system uses cabinet blasters in a segregated room. The room is built with one access door with a walk-through curtain. The blasting room is ventilated with an exhaust fan that provides a negative pressure. This assures general shop air is pulled into the room, preventing airborne debris from entering the shop area. The extraction fan(s) typically run only while the blast cabinets are running. This arrangement catches much more debris than standard practice, preventing it from drifting onto the cleaning technicians, previously cleaned parts, or into the shop area.

Access to the room is restricted to only one doorway with a curtain to maximize shop-to-room airflow. Note that some rooms have been built with three sides of walk-through curtains for maximum access. The extraction fans in these rooms, while substantial, could not create enough

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airflow through the curtains to control the airborne debris and dust still drifted throughout the shop.

3.0 Implementation Steps

- **Determine area in shop to be designated for blasting.**
 - Assure cleaning area has supporting utilities (compressed air for blasting, lighting, and electricity as needed).
 - Assure area has adequate space and access for equipment/supplies, and parts staging (outside of blasting room).
- **Research and choose blasting equipment**
 - Size cabinets, exhaust fans, and supply storage equipment, to needs.
 - Establish equipment usability/configuration to assure easy cleaning/maintenance and professional appearance.
 - Consider glass-bead as well as steel-shot machines. Suggest J-Belt tumbling design (as apposed to a rotating cage) for steel-shot type machines.
- **Establish new/adjusted repair/rebuild time requirement targets as needed.**
 - Eliminate the current processes/tooling/equipment strategy (only one best way).
- **Install the layout/equipment changes into the shop**
 - This may occur in phases, as needed, such as shop arrangement, then equipment final design/fabrication.
 - Promote change feedback as installation may surface new problems.
- **Train employees to use the new procedures/equipment. Follow-up/enforce immediately.**
 - Suggest parts are staged outside of Blast area. Transport parts cart to and from staging area immediately outside of blasting room.
- **Establish equipment maintenance schedule/responsibility.**
- **Review process/equipment performance once established.**
- **Establish and install any adjustments as needed.**

4.0 Benefits

Saves Labor – Well designed/organized cleaning equipment reduces cleaning efforts. Less effort is required to maintain a clean work area and fewer parts need follow-up cleaning before assembly.

Saves Time – Parts are not staged in different areas for different types of cleaning, reducing work flow interruptions. Protecting parts against airborne debris interferes less with cleaning.

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Increases Quality – More effective cleaning equipment simplifies cleaning processes and promotes high technician professionalism.

5.0 Resources Required

- Facility

- Re-organize work area to allow for blasting room, blasting cabinet(s) placement, incoming and outgoing staging, and supply storage as needed.

- Support Equipment

- Well-built blast cabinets, both manual and tumbling steel-shot cabinets.
- Effective walk-through doorway curtain.
- Sized-right exhaust system.
- Miscellaneous supply storage and work area cleaning tools (brooms, dust pans, etc.).

- People

- Establish current cleaning concerns with team
- Establish process, layout, and equipment improvements through team.

- Training

- Define/establish/enforce new processes, roles and responsibilities with production teams.
- Remove replaced items (general-area blast cabinets) from operations to prevent less effective cleaning.

6.0 Supporting Attachments / References

None.

7.0 Related Best Practices

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Parts Buffer Enhancement

8.0 Acknowledgements

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